

IN4MATX 232: Research in HCI

Class 2:
Research Contribution Types

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Participation

- Will make the class better
- I might call on some of you today, based on your Perusall comments
 - It's because I thought you had something productive to contribute!
- I have a bunch of points prepared, but I'd rather discuss what you have questions on

Multiple contribution styles

- Relatively common
 - Certainly empirical combines well with other styles
- But maybe less focused?
 - The more points you make, the more it dilutes the core takeaways of your work

Why are empirical and artifact contributions so prominent?

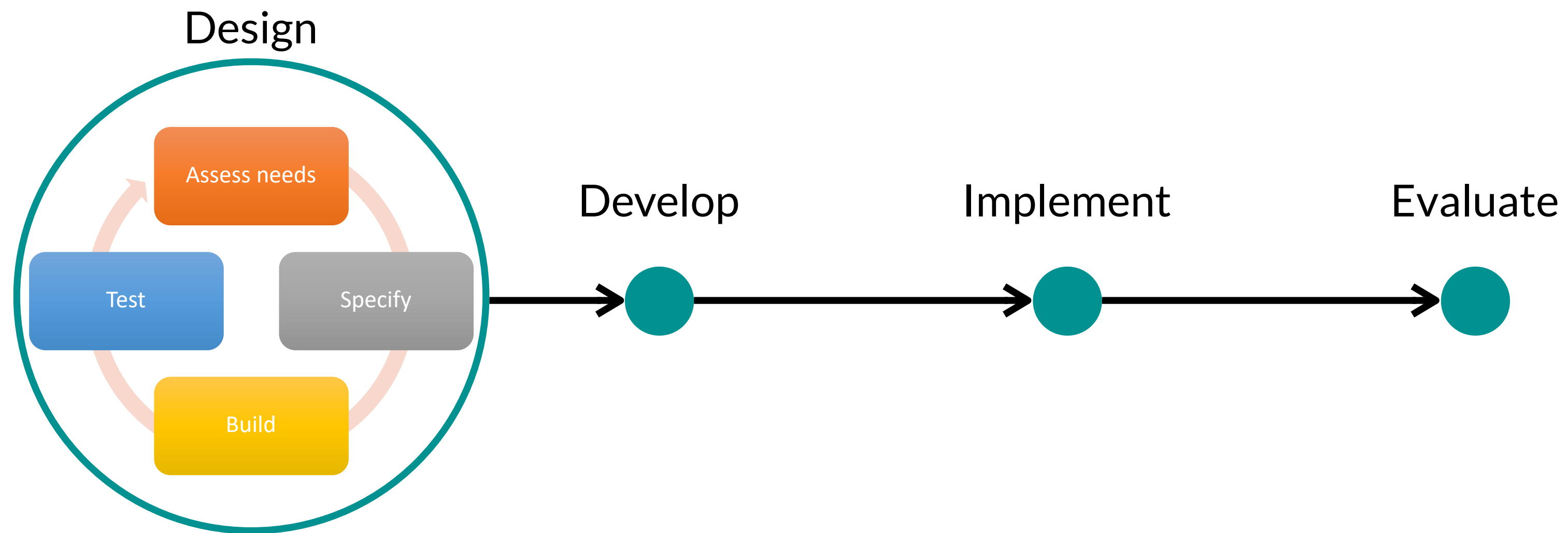
Applying Human-Centered Design

Information technology lifecycle



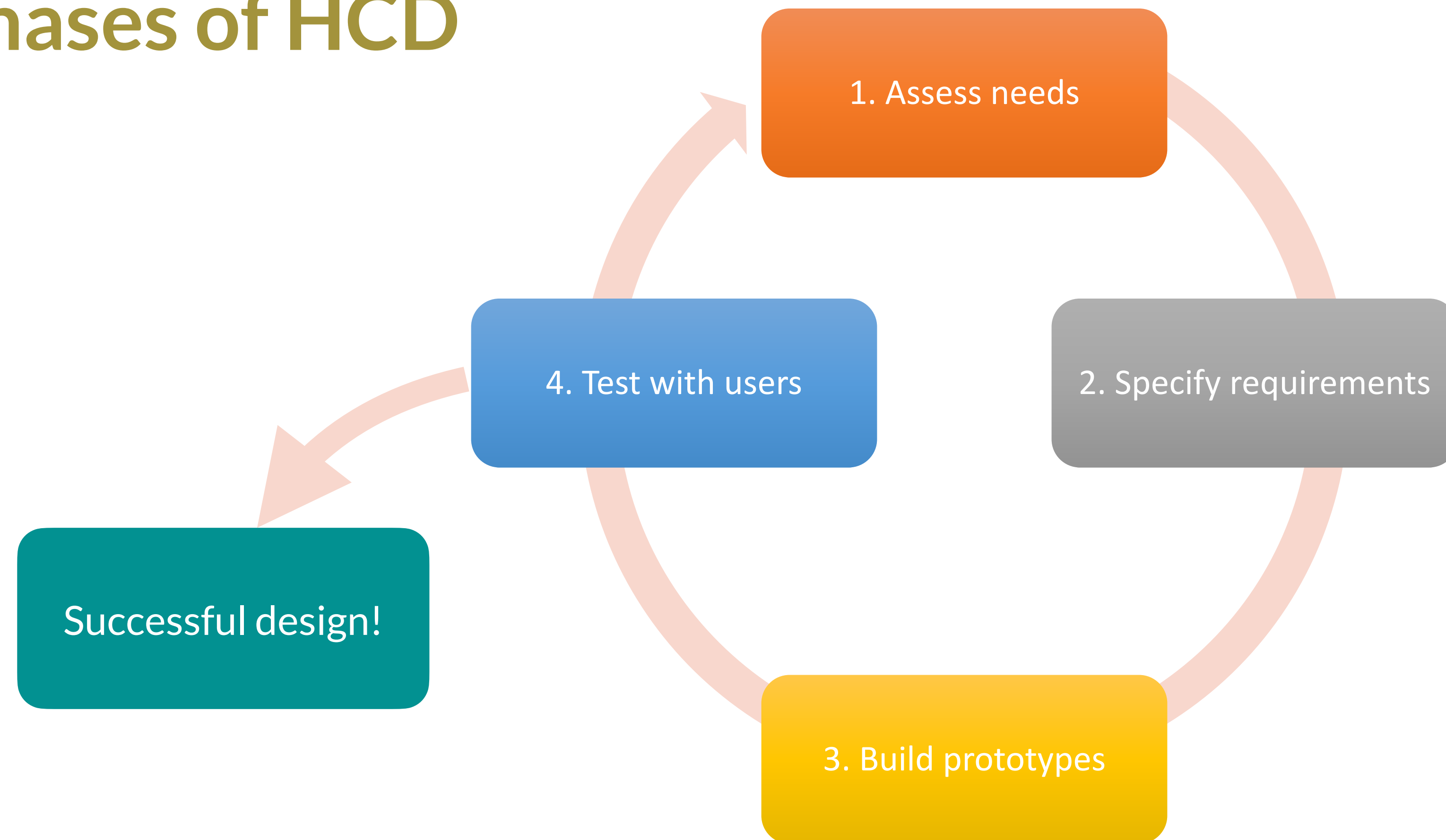
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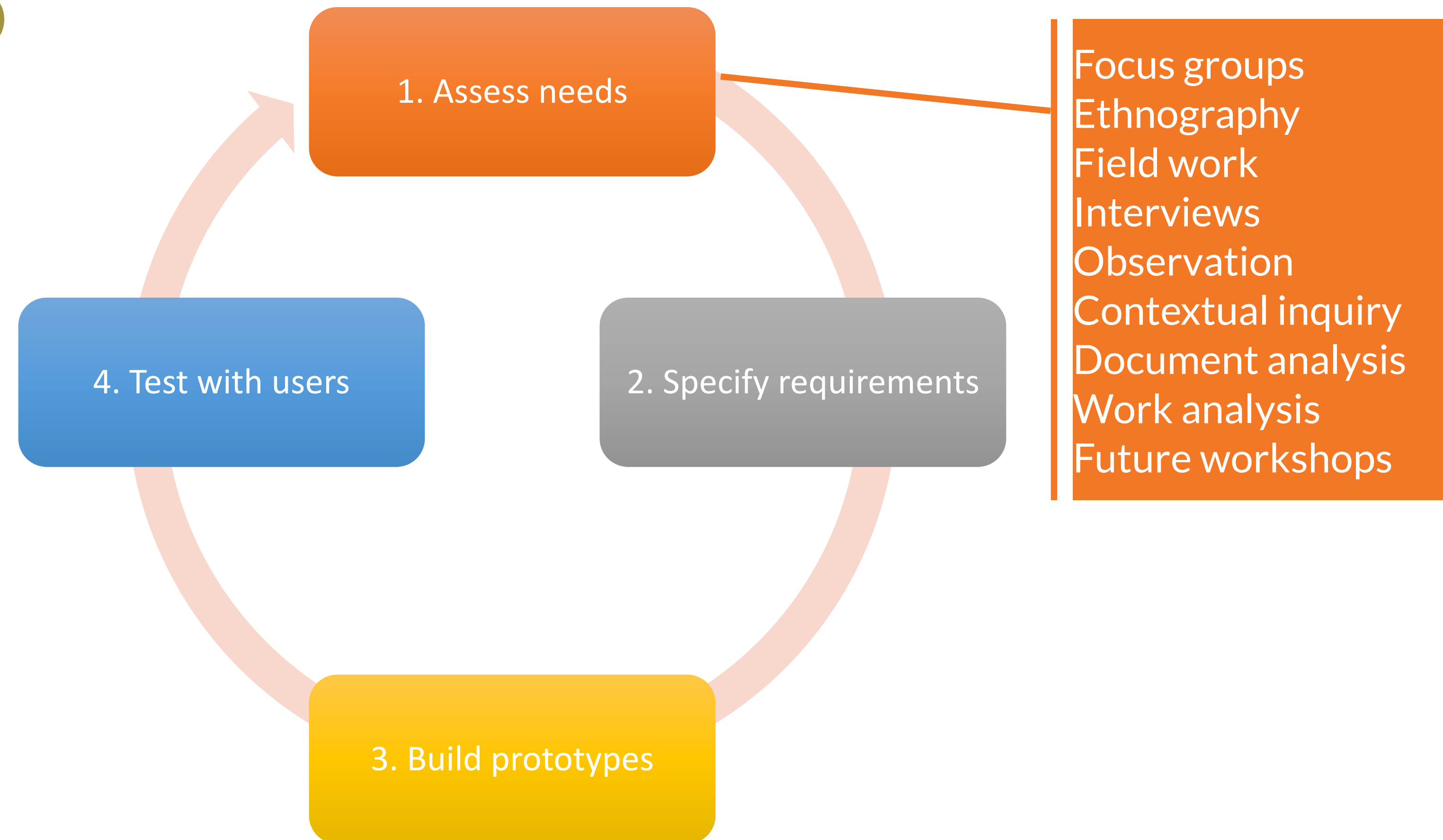
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Phases of HCD



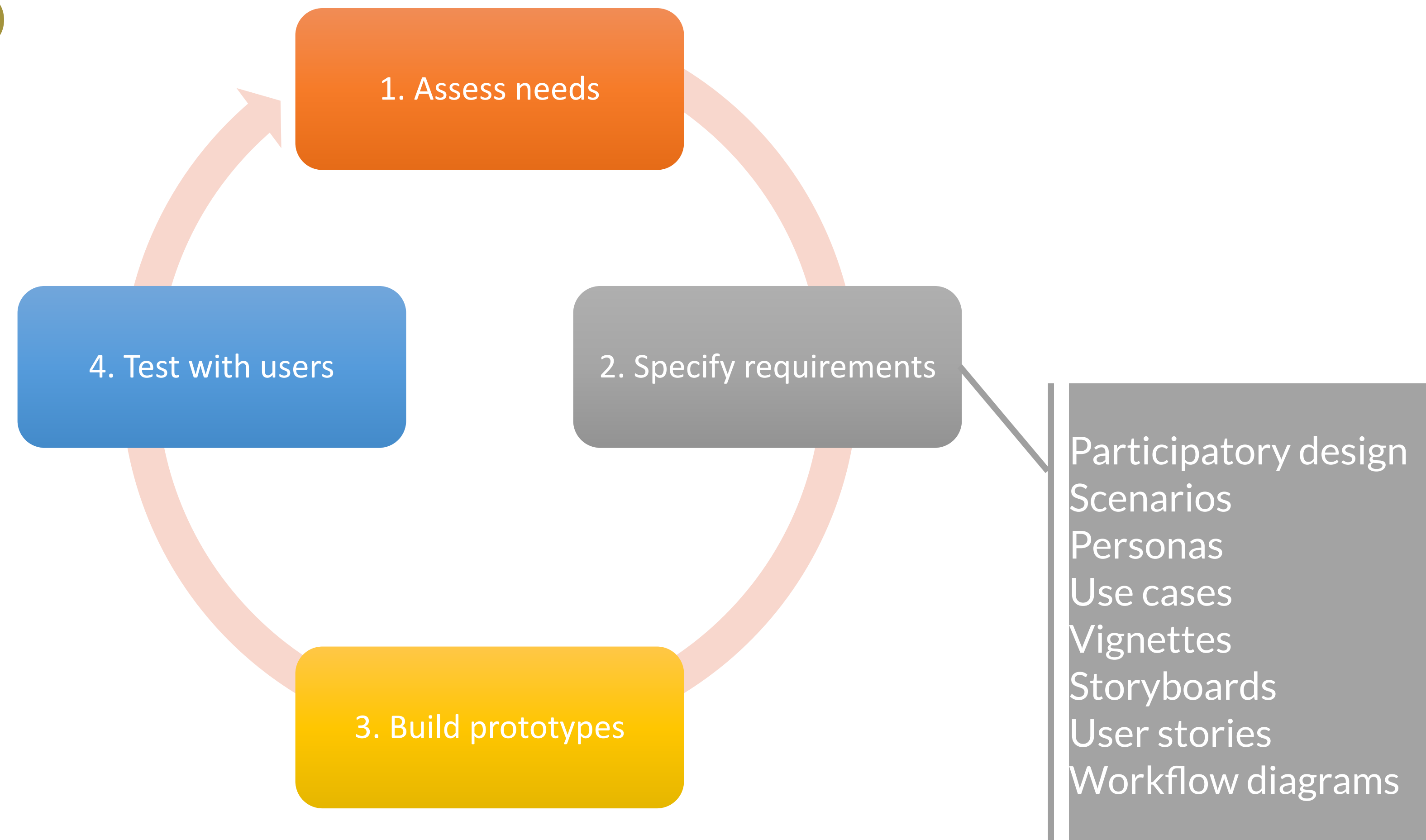
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Phases of HCD



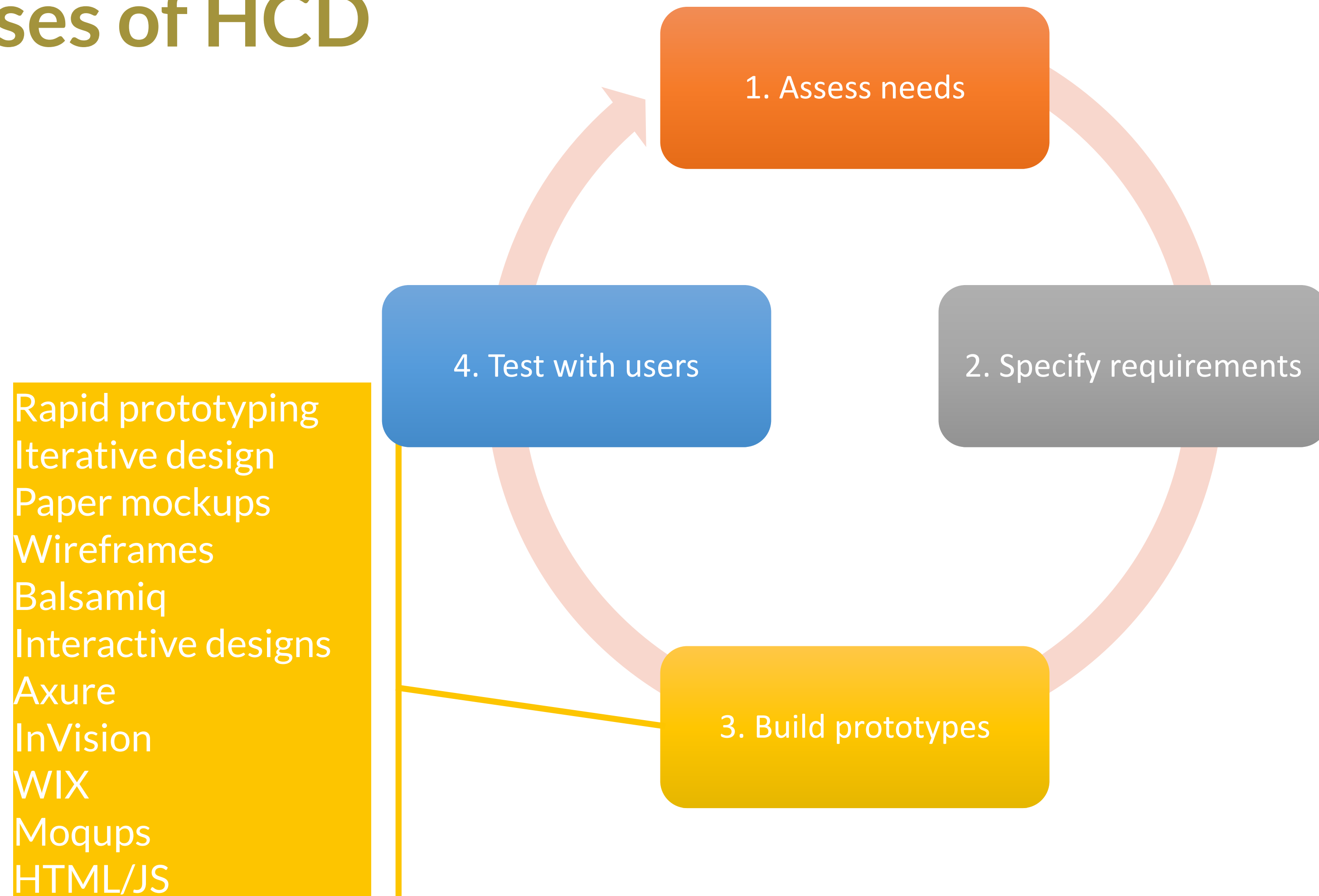
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Phases of HCD



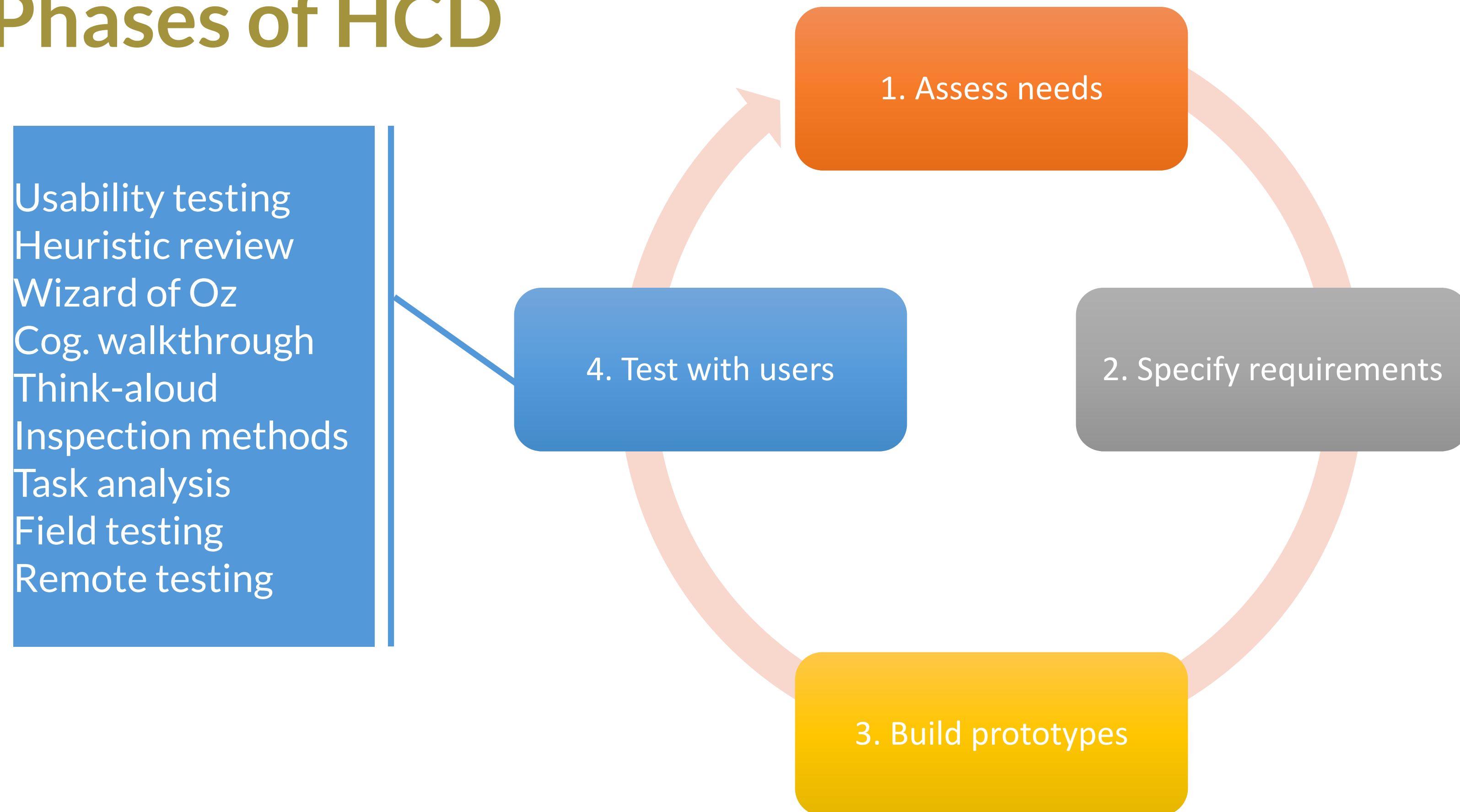
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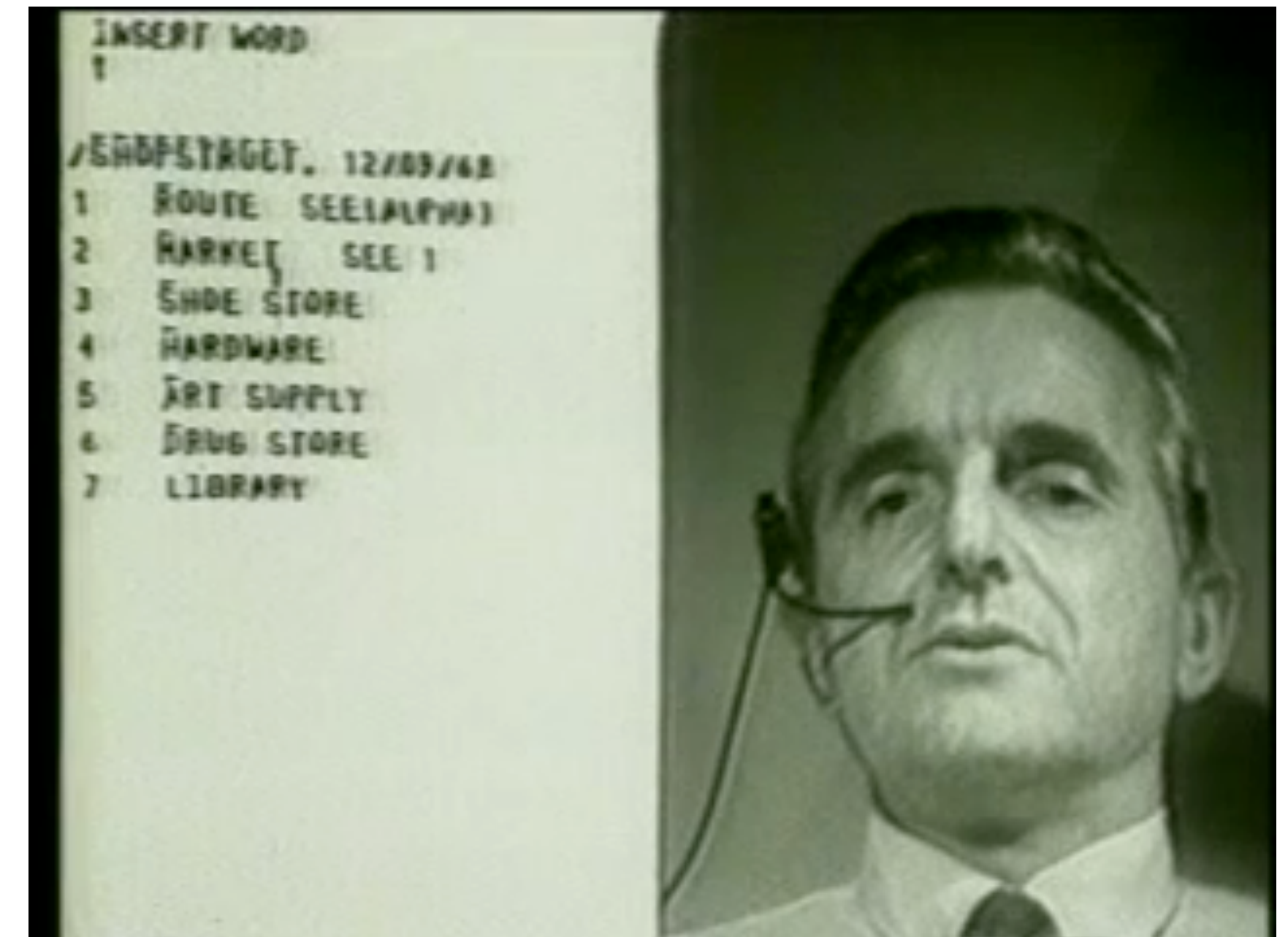
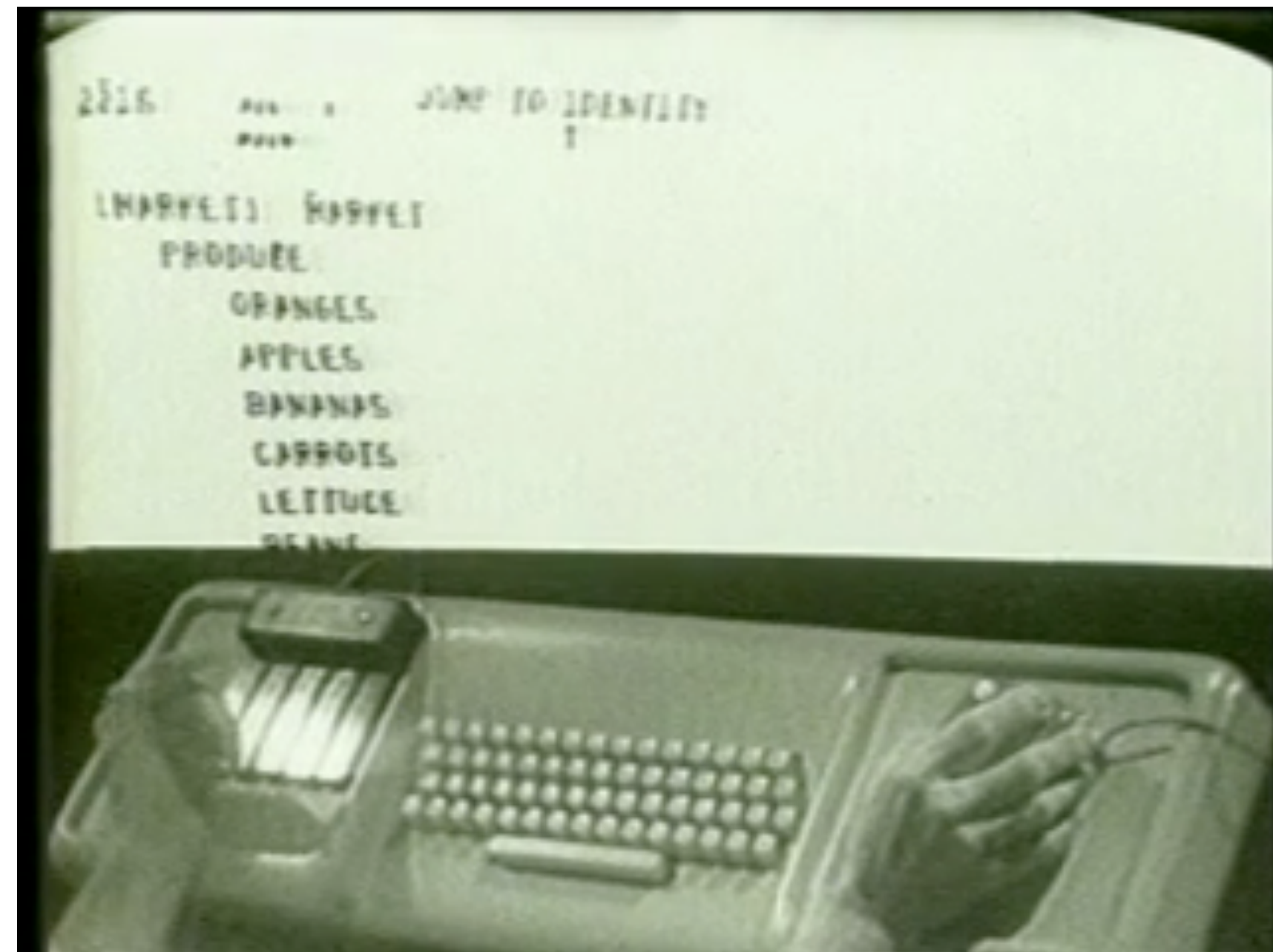
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Phases of HCD



**Why is evaluation not required
(and sometimes dissuaded) for artifact
contributions?**

Doug Engelbart's NLS (1968)



Empirical evaluation can take different forms

- Conducting an analysis on an existing dataset
- Giving a few examples of the solution in practice
 - “Here’s what my new artifact enabled us to make, which wouldn’t have been possible before”
- Demonstrating value on principles there’s a high degree of consensus on
 - Coding, like lines needed to do a task or execution time

What's “interesting” or “important”?

What's “interesting” or “important”?

- In the eye of the beholder
 - Translational work can be interesting, even if it's known in some other field
- Negative results (rejected hypotheses for quant work) can definitely be interesting

What's “interesting” or “important”?

- In my opinion, HCI equates “novel” and “interesting” in ways which undermine building up depth of knowledge
 - There's value in documenting things which some experts might know, but haven't articulated
 - We're not rewarded for more deeply understanding complex phenomena
 - Meta-analyses are extremely rare, though systematic reviews are frequent
- “Important” is a bit better, is influenced by societal conversations
 - But, “important” is an extremely high bar for any one paper

**What makes methods “sound” or data
“objective”?**

What makes methods “sound” or data “objective”?

- Varies a lot from method to method
- Emphasis on reproducibility, e.g., someone else should be able to recreate your method based on your description
- “Objective” is too high of a bar, particularly in a field as fast-moving as HCI
 - All studies/data have flaws, but the extent to which the flaws undermine the contribution can vary
 - But, we should aim to document the flaws

Methods *vs.* theory

Methods vs. theory

- Methods contributions aim to suggest how we should do our research and practice
 - How should we as researchers run our studies, develop our systems, share our opinions, etc.
 - How should UX practitioners run their studies...
- Theory mostly aims to produce knowledge (about people, systems, etc.)
 - Our methods may be informed by theories (e.g., Activity Theory as an analytic lens, designing a system which uses principles from Social Cognitive Theory)

Theory

- “Should be testable and falsifiable”
 - Is this statement too quant-centric?
 - I’d actually argue no, you should be able to articulate whether/how qualitatively-derived theories apply in your new context/circumstances
 - The evaluation might not be as binary as this statement suggests, but it does occur

Datasets vs. empirical

Datasets vs. empirical

- Empirical studies will tell us something about how people perceive/use technology, etc.
 - Data (qualitative, quantitative, both) will be collected
 - But the data may or may not be shared with the research community (protecting anonymity, lack of utility)
- For datasets, the released data *is* the primary contribution
 - A large data repository that multiple researchers would find useful, for example
 - The dataset might not provide a study which uses it, and instead expects future researchers to instead

Datasets

Rico: A Mobile App Dataset for Building Data-Driven Design Applications

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ABSTRACT

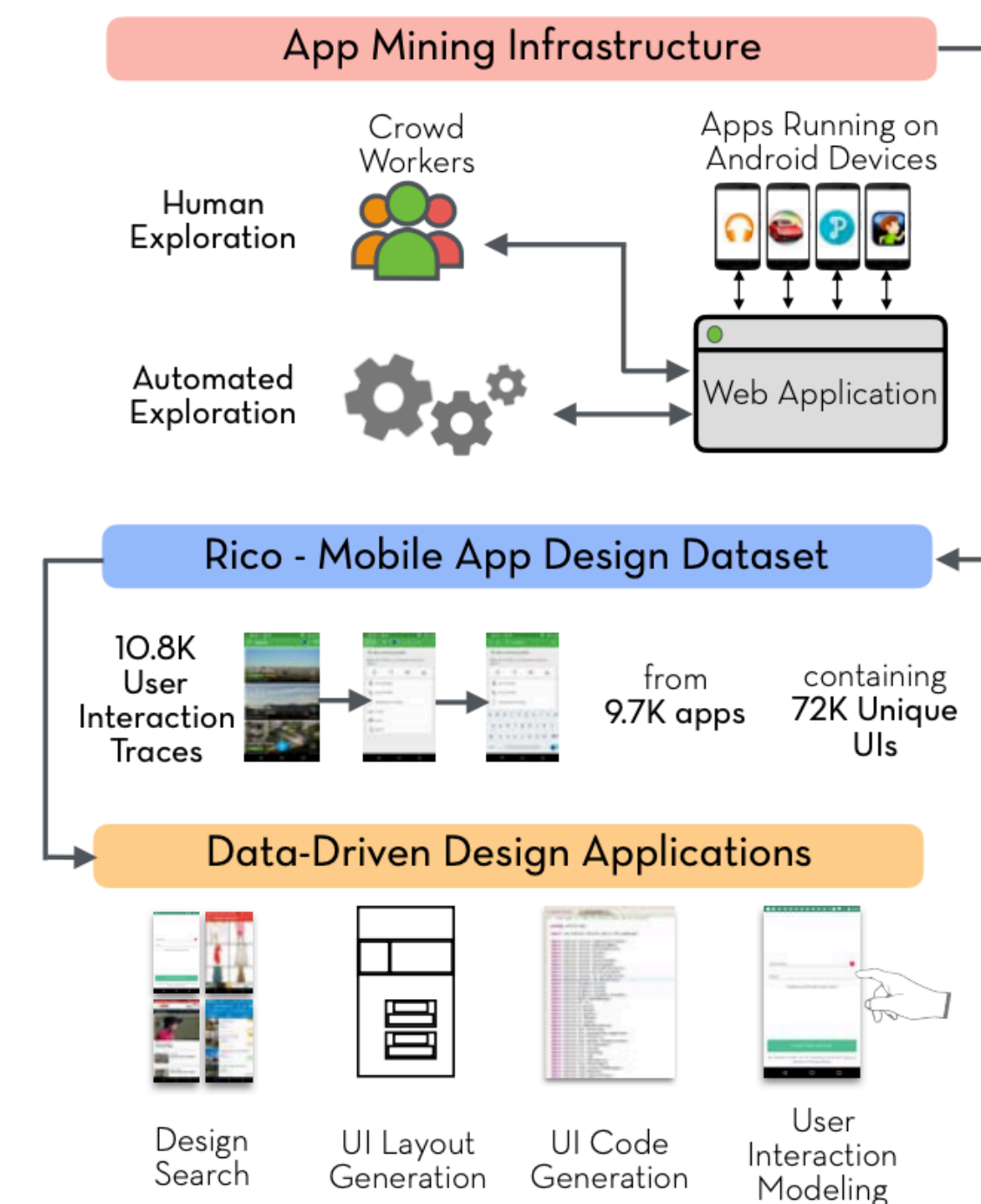
Data-driven models help mobile app designers understand best practices and trends, and can be used to make predictions about design performance and support the creation of adaptive UIs. This paper presents Rico, the largest repository of mobile app designs to date, created to support five classes of data-driven applications: design search, UI layout generation, UI code generation, user interaction modeling, and user perception prediction. To create Rico, we built a system that combines crowdsourcing and automation to scalably mine design and interaction data from Android apps at runtime. The Rico dataset contains design data from more than 9.7k Android apps spanning 27 categories. It exposes visual, textual, structural, and interactive design properties of more than 72k unique UI screens. To demonstrate the kinds of applications that Rico enables, we present results from training an autoencoder for UI layout similarity, which supports query-by-example search over UIs.

ACM Classification Keywords

D.2.2 Software Engineering: Design Tools and Techniques

Author Keywords

Mobile app design; design mining; design search; app datasets



<https://dl.acm.org/doi/abs/10.1145/3126594.3126651>

Meta-analyses, Systematic analyses, and Survey contributions

Meta, Systematic, Survey

- Meta-analysis: considers a *question* that people have repeatedly studied
 - Settings may have changed—different populations, for example
 - Often quantitative in nature. Is there an effect? If so, how much?
- Systematic review: considers a *topic* that people have repeatedly studied
 - Tries to build up what the field has learned across disjointed settings
 - Tries to identify underexplored research areas or questions

Meta, Systematic, Survey

- We don't do meta-analyses in HCI all that often
 - Why?

Assorted thoughts

Assorted thoughts

- Contribution type is implicitly associated with acceptance rate, whether it should be or not
 - Authors are more used to making empirical/artifact contributions, so it can be harder to make a “good” opinion/survey/dataset contribution
 - Reviewers are also more used to evaluating empirical/artifact contributions, and may be more critical in evaluations of others
- Opinions, surveys, etc. tend to, be led by more senior community members
 - They require perspective and understanding of the research field
 - But, I have seen many students lead them!

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